BMI500 HomeWork 7 (Lecture 7) Hypothesis

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Question number: 2

Hypothesis: Reducing the agents' movements while holding other factors (population, transmission probability, etc.) constant, I hypothesize the maximum number of infected agents (height) to be lowered and the timing of the infection peak to be delayed. Less movement should limit the spread of infected agents and the transmission of the epidemic to susceptible agents.

I expect the individual simulations to slightly vary with general pattern being similar. The infection curve should rise over time, peaks, and declines eventually. The specific height and the timing of the infection peak should slightly vary across simulations because the agents' initial positions and movement are stochastic. This assumption holds for both normal agents and agents with reduced movement.

I expect the average agent-based curves to largely resemble the broad trends of a SIR model in the lecture. Assuming we are only considering susceptible, infected, and recovered curves, the susceptible population should rapidly decrease and slightly increase, the infected population should rise and then decline, and recovered population should increase over time. I would not expect curve locations to be exactly identical given randomness is involved and the sample size is relatively small.